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Ascertainment of non-coding genes as key molecular players in cervical squamous cell carcinoma through the systems biology approach

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Introduction

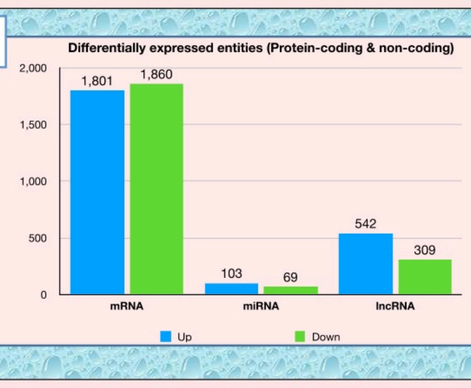
Cancer is an extremely complex, heterogeneous disease. In India, as per the data from ICMR, lung cancer, colorectal cancer in men and breast cancer, cervical cancer (CC) in women are the most frequent (6-29% of all cancer cases) and CC causes more deaths in Indian women than in any other country. Even though the prevalence of CC can in theory be diminished using HPV vaccines (a large fraction of CC is caused by the HPV virus), there is still need to understand the biological make up of different sub-types of CC, to identify potential vulnerabilities (drug-targets) for future personalized medicine approaches.

Aims & Objectives

To identify the novel non-coding elements that has a role in cervical cancer progression, we explored an integrative network approach by combining differential expression analysis, WGCNA, PPI network and regulatory networks.

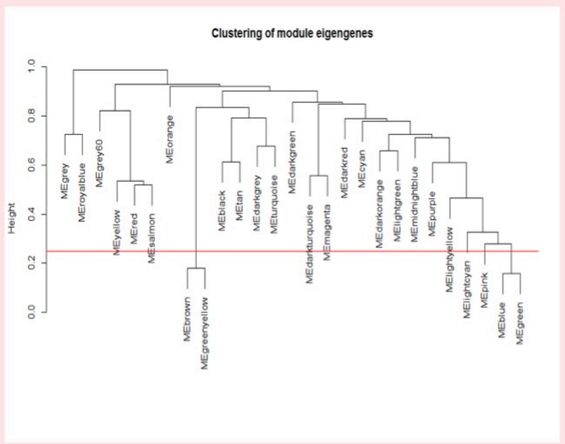
1. Screening significantly differentially expressed genes (protein-coding and non-coding)
2. Gene co-expression network construction and analysis
3. Protein-protein interaction network
4. Regulatory network construction and analysis

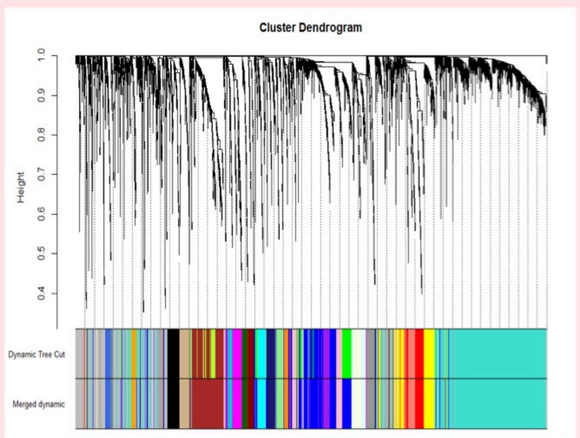
Differential expression analysis

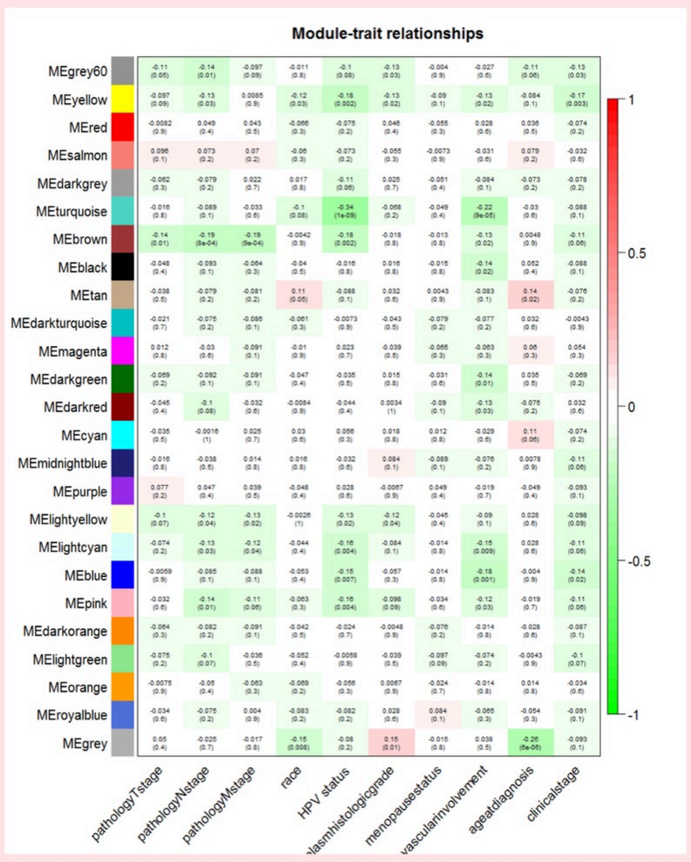


|log2FC| > 2
pvalue < 0.01

Weighted gene co-expression network construction & analysis







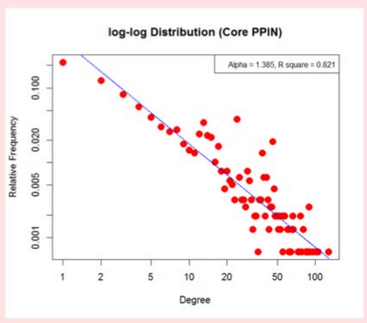
Soft-thresholding power ß : 5
Minimum module size :30
Module merging threshold: 0.25
No. of detected modules: 27
Merged modules: 2

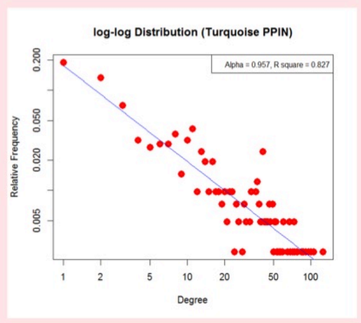
Highly correlated modules with HPV status

Module	r	pvalue
Turquoise	-0.34	1e-09
Yellow	-0.18	0.002
Brown	-0.18	0.002
Blue	-0.15	0.007
Lightcyan	-0.16	0.004
Pink	-0.16	0.004
Lightyellow	-0.13	0.02

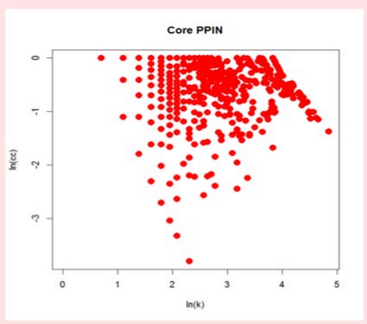
Protein-protein interaction network analysis

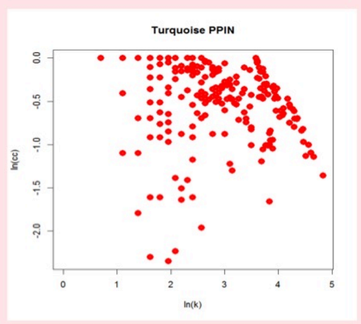
Topological properties of networks:



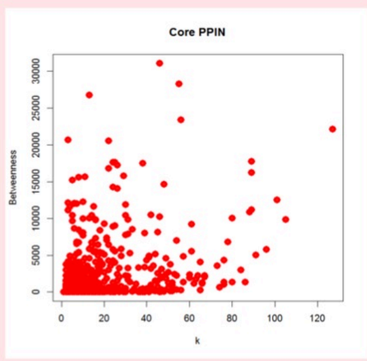


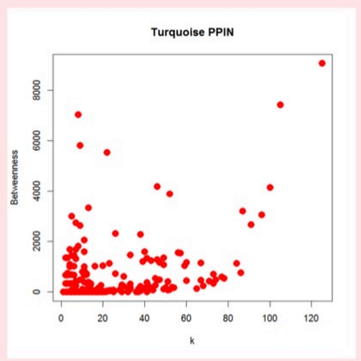
Left panel: Degree distribution of core or primary network and Turquoise module networks follow the property of power law and scale free nature.





Left panel: Degree and clustering coefficient (CC) of both core and turquoise module networks are negatively correlated.





Left panel: Degree and Betweenness correlations of core and turquoise module networks are positively correlated.

Regulatory & ceRNA network analysis

Approach

- Cancer-specific regulatory networks built from TCGA cervical cancer transcriptomes: DElncRNA → target DEmiRNA and DEmiRNA → hub DEmRNA networks, combined into a lncRNA–miRNA–mRNA ceRNA network.
- Hub genes of the dysregulated PPI network validated with a support vector machine (SVM) model.

Key findings

- KCNQ1OT1 is the hub lncRNA of the lncRNA–miRNA network, with a regulatory role in transcription of target genes via epigenetic modifications.
- hsa-miR-184 is present in both target networks and is implicated in several cancer types.
- hsa-miR-637 is the hub microRNA of the miRNA–mRNA network, acting in post-transcriptional regulation.
- Key genes are enriched for cell division, mitotic nuclear division and cell-cycle checkpoints (GO) and the cell cycle (KEGG).
- Survival analysis assessed the prognostic value of the candidates for overall survival in cervical cancer patients.

Key Conclusion:

Our findings provide key insights into the regulatory role of a novel lncRNA KCNQ1OT1 via cancer-specific expression networks in the pathogenesis of cervical cancer.

It plays an important regulatory role in transcription of target genes via epigenetic modifications

Acknowledgements

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Contact me: